

Data Analysis in Power BI

Course Introduction

Data Analysis with Power BI

Learning Objectives

01	Import multiple sources of data into Power BI with the Power Query Editor.	02	Create a report data model using queries as the model tables.
03	Analyze the data model with DAX Measures & Calculated Columns.	04	Build a Power BI report containing a variety of different visuals.



Course Instructor

Joseph Yeates

About Joseph

Joseph is passionate about driving better decision-making by creating, curating, and communicating insights from data.



Experience

A data enthusiast with a background in Retail Banking, he is an advocate for making advanced analytics more accessible to a wider audience by integrating them into business intelligence solutions.

Joseph helps organize the Vancouver Power BI & Modern Excel User Group. An active speaker, he has presented sessions on combining data science with business intelligence at conferences in the Pacific Northwest and across North America.

Power BI Overview

Power BI Overview

Power BI is an all-encompassing Business Intelligence tool.

It is possible to conduct all the main tasks and skills for any data reporting within Power BI.



Data Transformation

Data Modeling

Data Analysis

Data Visualization

User Experience

Sharing

Power BI Overview

Power BI is an all-encompassing Business Intelligence tool.

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Data Transformation

Automate repetitive and manual data manipulation

Power Query Editor

Data Modeling

Data Analysis

Data Visualization

User Experience

Sharing

Power BI Overview

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Data Transformation



Data Modeling

Provides the data source within the report to build visuals and report elements

Power Pivot - Model View

Data Analysis

Data Visualization

User Experience

Sharing

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Data Transformation

Data Modeling



Data Analysis

Analyze data model and summarize data using DAX measures

Power BI Model &
Data View

Data Visualization

User Experience

Sharing

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Data Transformation

Data Modeling

Data Analysis



Data
Visualization

Storytelling with data;
creating dashboards
and visual reports

Power BI Report View

User Experience

Sharing

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Data Transformation

Data Modeling

Data Analysis

Data Visualization



User
Experience

User Experience
elements & interactivity

Power BI Report View

Sharing

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Data Transformation

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Data Visualization

User Experience



Sharing

Share report with others and start to collaborate

Power BI Service

Power BI Overview

We will focus on the first four areas of Power BI within this course.

Power BI is more than a data visualization tool.

It is a market-leading business intelligence tool, which is why this course covers **ALL** areas with a focus on creating visuals and user experience elements.



Data Transformation

Data Modeling

Data Analysis

Data Visualization

User Experience

Sharing

Data Sets

Retail Sales Fact Table

ID	Store ID	Category ID	Date	Sales \$ USD
105	1	7	01/01/2022	400

Date Dimension

Date	Year	Quarter	Month	Week
01/01/2022	2022	1	1	1

Store Dimension

Store ID	Store Type	Store Size (sqft)	Location
1	CORE	151,315	Kansas

Category Dimension

Category ID	Department	Product Group
7	Clothing	Sports

Retail Cost Fact Table

Store ID	Date	Wages	Rent	Target Sales
7	01/01/2022	#,###.00	#,###.00	#,###.00

Power Query

Chapter Introduction – Power Query

We will connect to the source data for our Power BI report — importing and transforming it in the Power Query Editor before loading it into our report.



How to connect to different data sources



How to manage our queries in the Editor, following a few best practices



How to perform some common data transformations to reshape our source data



Review a few advanced features



How to add additional columns of data



Finally, loading our queries into the report Data Model

Why Power Query

Power Query is a foundational part of Power BI.



Explore more in the *Power Query Fundamentals* course by CFI



Overview

Core ETL (Extract, Transform, Load) engine of Power BI

Automates repetitive/manual data transformations

User-friendly interface for creating documented, repeatable steps



Characteristics

Transformation steps vary by data source and report

Many functions available beyond those demonstrated



Best Practices

Understand your data and reporting goals

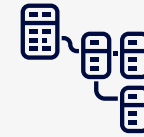
Use transformations to bridge gaps between raw and desired data

Problem-solving is essential for applying these concepts

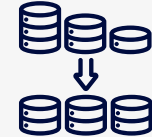
Data Modeling & Basic DAX Measures

Chapter Introduction – Data Modeling & Basic DAX Measures

We will take the queries that we created in the Power Query Editor and relate them together to create our report Data Model.



Data Modeling



Database
Normalization



Fact & Dimension,
STAR Schema

DAX

Data Analysis
Expressions

Chapter Introduction – Data Modeling & Basic DAX Measures



For a more comprehensive view,
please check out the *Data
Modeling & DAX* courses by CFI



Updating the aggregation types across our report



Updating the formatting of aggregations across our report



And structuring our data so that it is stored efficiently, with
predictable behavior and no errors

Data Modeling



It transforms data into a form that is easier to understand and quick to answer our reporting questions.



Each data model will be different. It should reflect the state of the business it is analyzing and the report's needs.



However, there are several common aspects across data models.

Database Normalization

Database normalization is a method of storing and structuring data efficiently for storage and analysis.

It has four key benefits:

 Protects data integrity

 Reduces storage space

 Easier to maintain

 Improves query speed

Can you see any problems with the layout of the data?

Product #	Product Group	Product Color	01/31/2019	05/14/2019	05/21/2019	07/13/2019
80725	Tools	Chrome	192	0	0	0
80726	Electric Doors	White	0	1,400	0	0
80726	Electric Doors	White	0	0	1,300	0
90724	Radios	Black	0	0	0	57

Database Normalization

-  All the attributes in this table should describe a sale. They should NOT describe the product.
-  Rows in the table should be uniquely identified.
-  Similar data should be in the same column – not spread out across multiple columns.

Can you see any problems with the layout of the data?

Product #	Product Group	Product Color	01/31/2019	05/14/2019	05/21/2019	07/13/2019
80725	Tools	Chrome	192	0	0	0
80726	Electric Doors	White	0	1,400	0	0
80726	Electric Doors	White	0	0	1,300	0
90724	Radios	Black	0	0	0	57

Fact & Dimension Tables

A **fact table** is a table that records a particular event or measurement.

Dimension tables add a description to the attributes in fact tables.

Retail Sales Fact Table

Product #	Store	Date	Sales
80725	307	01/31/2019	192
80726	308	05/14/2019	1,400
80726	309	05/21/2019	1,300
90724	310	07/13/2019	57

Fact & Dimension Tables

A **fact table** is a table that records a particular event or measurement.

Dimension tables add a description to the attributes in fact tables.

Product Dimension Table

Product #	Department	Group	Color
80725	Electronics	Doors	White
80726	Clothing	Accessories	Black
90724	Kitchen	Cutlery	Silver

Date Dimension Table

Date	Year	Month
01/31/2019	2019	January
05/14/2019	2019	May
05/21/2019	2019	May
07/13/2019	2019	July

Fact & Dimension Tables

A **fact table** is a table that records a particular event or measurement.

Dimension tables add a description to the attributes in fact tables.



Sales fact tables contain the data that we measure about our business.



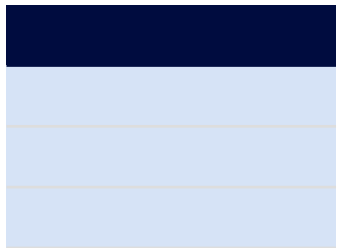
Dimension tables add descriptions to the attributes in our sales fact table.



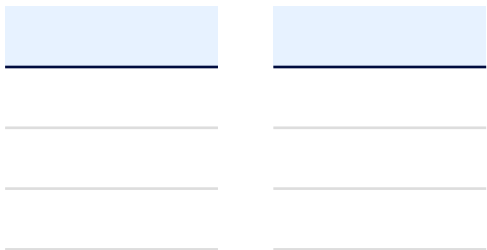
The fact table and the dimension tables share common columns: Product Number and Date.

Relate Fact & Dimension Tables

A **fact table** is a table that records a particular event or measurement.

A diagram showing a fact table structure. It consists of a dark blue header row and three light blue data rows, all contained within a light blue rectangular frame.

Dimension tables add a description to the attributes in fact tables.

A diagram showing two dimension table structures. Each structure consists of a light blue header row and three light blue data rows, all contained within a light blue rectangular frame.

Product Dimension Table

Product #	Department	Group	Color
80725	Electronics	Doors	White
80726	Clothing	Accessories	Black
90724	Kitchen	Cutlery	Silver



When the Product Number appears on the Product Dimension, it is a unique identifier. It can only appear once in the dimension table and is considered a **primary key**.

Relate Fact & Dimension Tables

A **fact table** is a table that records a particular event or measurement.

Dimension tables add a description to the attributes in fact tables.

Retail Sales Fact Table

Product #	Store	Date	Sales
80725	307	01/31/2019	192
80726	308	05/14/2019	1,400
80726	309	05/21/2019	1,300
90724	310	07/13/2019	57



In the Sales Fact Table, the Product Number is a **foreign key** that links the fact table to details in the dimension table. It can appear multiple times

Relate Fact & Dimension Tables

A **fact table** is a table that records a particular event or measurement.

Dimension tables add a description to the attributes in fact tables.

Retail Sales Fact Table

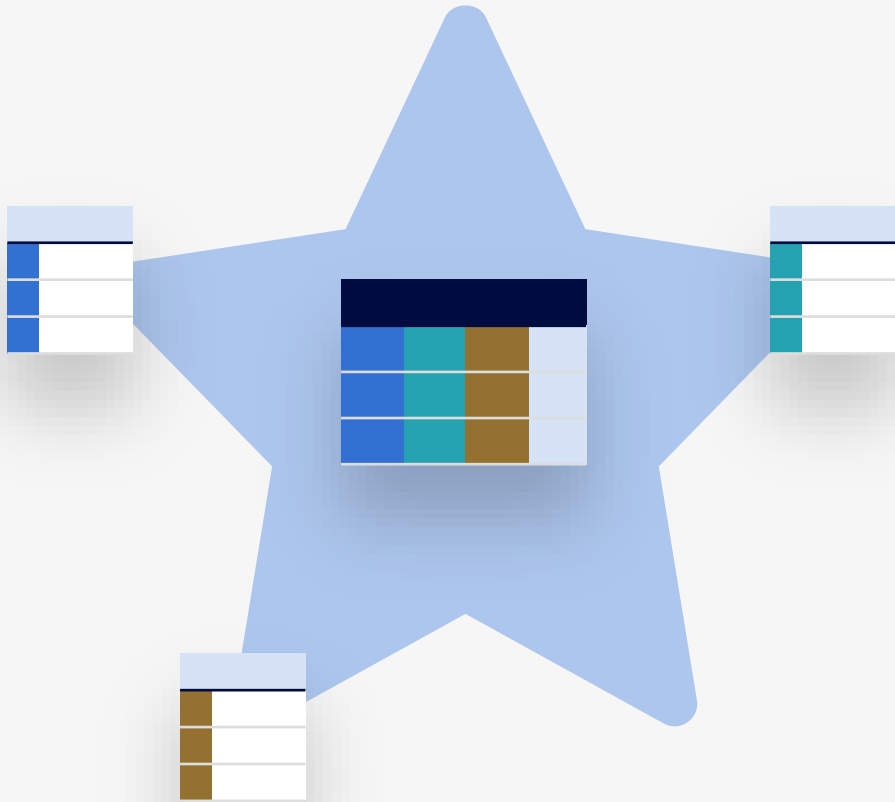
Product #	Store	Date	Sales
80725	307	01/31/2019	192
80726	308	05/14/2019	1,400
80726	309	05/21/2019	1,300
90724	310	07/13/2019	57

Date Dimension Table

Date	Year	Month
01/31/2019	2019	January
05/14/2019	2019	May
05/21/2019	2019	May
07/13/2019	2019	July

Relate Fact & Dimension Tables – **Star Schema**

A single fact table, surrounded by multiple dimension tables, is referred to as a **Star Schema data model**.



1-M relationships

- Primary Keys appear only once in the dimension tables.
- The related Foreign Keys can appear many times in the fact table.

Organize data with Power BI

- Power BI provides ability to database normalization.
- This foundation is to build our reports in Power BI, so that they are efficient, perform well, and have no data integrity issues.

Date Dimension



Defining our own date dimension can give us more control over our data model.



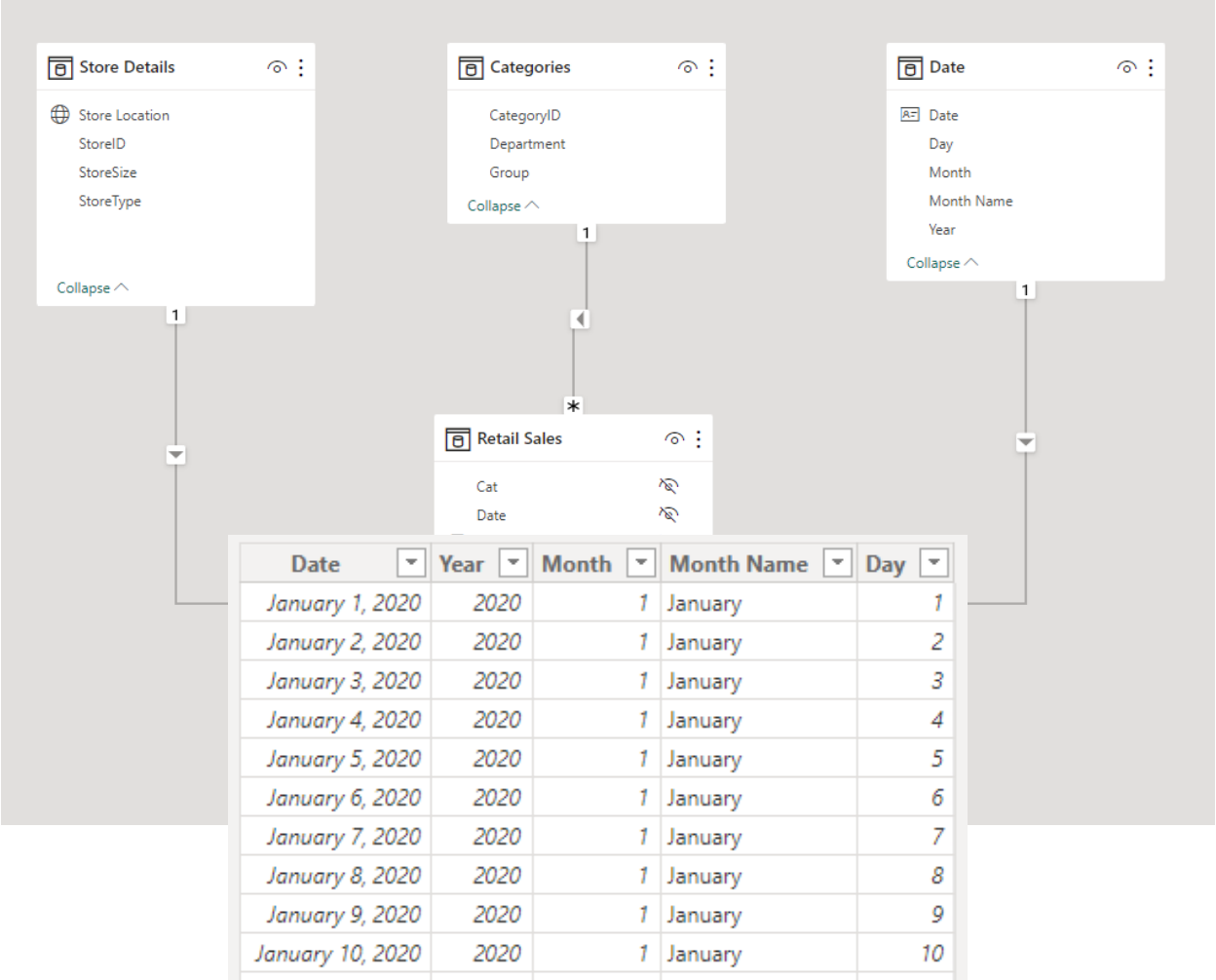
We can ensure that our report performs efficiently and as expected.



A Date Dimension was added to our data model and marked as a date table.



Dates in a date dimension table must be consecutive.



DAX – Data Analysis Expressions



DAX is used to write formulas in Power BI.



These formulas can help us create, aggregate, or summarize data in our reports.



DAX formulas can be categorized into one of two groups: Calculated Columns and Measures.

DAX Measures

When we are creating business intelligence reports, there is a need to summarize our data.



Seeing data as a metric.



Define an aggregation type and calculate the Total amount for a specific column.



This single measure can be used to populate visuals and tables with Total Sales.

Retail Sales Fact Table

Product #	Store	Date	Sales
80725	307	01/31/2019	192
80726	308	05/14/2019	1,400
80726	309	05/21/2019	1,300
90724	310	07/13/2019	57

Total Sales = SUM('Retail Sales'[Sales])
= 5,916.34

Sales Page – Visuals & DAX

Chapter Introduction

We will start to visualize our data – leveraging our work importing, transforming, and modeling our data.



How we choose to visualize our data and plan our report pages is crucial to any analyst working in Power BI.



Cover the theory of how we can create visuals using DAX measures.



Create a variety of different types of visuals.



See how we can create further DAX measures to fit the need for particular analysis and visualizations.



Filter the data in our report to change the context of our visuals.

Visualizing with Context

Total Sales
5,916.34



Context is one of the most fundamental concepts behind DAX measures.

Retail Sales Fact Table

Date	Department	Group	Store ID	Sales
June 30, 2019	Kitchen	Cutlery	1	1,051.30
March 31, 2019	Kitchen	Cutlery	2	1,091.54
September 1, 2019	Kitchen	Gadgets	2	1,816.00
December 1, 2019	Kitchen	Gadgets	6	1,957.50

Store ID	Total Sales
1	1,051.30
2	2,907.54
6	1,957.50
Total	5,916.34

Group	Total Sales
Cutlery	2,142.84
Gadgets	3,773.50
Total	5,916.34

DAX Calculated Columns

Retail Sales Fact Table

Date	Department	Group	Sales	Group & Dept
June 30, 2019	Kitchen	Cutlery	1,051.30	Cutlery: Kitchen
March 31, 2019	Kitchen	Cutlery	1,091.54	Cutlery: Kitchen
September 1, 2019	Kitchen	Gadgets	1,816.00	Gadgets: Kitchen
December 1, 2019	Kitchen	Gadgets	1,957.50	Gadgets: Kitchen

Group & Dept = [Group] & ": " & [Department]

Calculated Columns return a result for each individual row in our table.

Cost – Visuals & DAX

Chapter Introduction

**An additional fact table
containing information on the
business's costs and the sales
targets**



Comparing the sales to target values



How well the overall business, as well as individual stores, are performing

We will create a second report page with two main objectives:

-  Analyzing the two main costs (wages and rent) by store
-  Tracking a KPI – sales vs target over time



An additional fact table can help demonstrate the possibilities that we have in modeling data and the importance of understanding the data available to us.

Course Summary

Course Summary

We've covered a lot of different topics in all areas of Power BI:



Imported data into the Power Query Editor – to automate our data transformations



Defined a variety of DAX measures and calculated columns to analyze the data in our data model



Modeled our queries into a Star Schema data model



Created two pages of report visuals to analyze Retail Sales & Retail Cost & KPI Target Tracking



We can use these skills to meet more advanced requirements and better communicate key insights with our Power BI reports.